

Use Case View Product Detail (Customer / Product Manager)

1. Use case code

- UC235

2. Brief Description

This use case describes how a Customer or Product Manager views the detailed information of a selected product in AIMS. After a product is selected, the client calls the product-detail API with the product id. The system validates the id, retrieves the product and its related records from the database, normalizes the product status, and maps the data into a product-detail response containing the general information and the type-specific information. The client then renders the detail page (customer detail page or manager view). The product-detail data endpoint (GET /products/:id) is read-only and public; role-based access control is enforced only on the Product Manager management routes on the client side, not on the data endpoint itself.

3. Actors

- Customer
- Product Manager

4. Preconditions

- The product id provided in the request is a positive-integer string (matching the pattern `^[1-9]d*$`); otherwise the request is rejected before any lookup.
- For Customer: the customer reaches the detail page through the public catalog route /product-catalog/:id (no authentication required).
- For Product Manager: the user is authenticated and has the 'Product Manager' role, and reaches the detail through the protected route /manager/products/view/:id.
- The product category is expected to be one of the supported types (Book, Newspaper, CD, DVD). Note: in the current code an unrecognized category is normalized to BOOK rather than being rejected.

5. Basic Flow of Events

1. The actor selects a product from the homepage, product list, search result, or product management screen.
2. The client sends a request to GET /products/:id with the selected product id.
3. The system validates the product id format (it must be a positive integer).
4. The system retrieves the product together with its related records (printable product → book / newspaper, disc product → CD / DVD, and book authors) from the database.
5. If the product is not found, the system returns a 404 response; otherwise it normalizes the product status and maps the record into the product-detail response, including the type-specific information.
6. The client displays the product detail (customer detail page or manager view), showing the general details and the type-specific details.
7. The actor reads the information and may continue browsing, return to the previous page, add the product to the cart (Customer), or perform management actions (Product Manager).

6. Alternative Flows

No	Location	Condition	Action	Resume location
1	At step 3	The product id is not a positive-integer string (fails the <code>^[1-9]\d*\$</code> check).	The API returns 400 Bad Request with message “Invalid product identifier”.	Use case ends
2	At step 5	The selected product does not exist in the database.	The API returns 404 Not Found (“Product {id} not found”); the customer detail page shows “Product not found.” and lets the actor return or keep browsing.	Use case ends
3	At step 4	A database or service error occurs while reading the product.	The error propagates as an HTTP 500 response (this use case does not write a dedicated server-side error log); the client shows a generic message “Cannot load product detail from backend.” and allows the actor to reload.	Use case ends
4	Before step 6 (Product Manager management route only)	The user opening the manager view is not authenticated or does not have the 'Product Manager' role.	The client route guards (authGuard / roleGuard) block access and redirect the user to login.	Use case ends

5	At step 6	The product status is unavailable (DEACTIVATED, UNAVAILABLE, or OUT_OF_STOCK).	For Customer: the page shows an availability notice and disables the quantity selector and the Add-to-Cart button. For Product Manager: the manager view still shows the full detail together with the status.	Continue at step 6 (detail is still displayed)
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7. Input Data

No	Attribute	Meaning	Format	Sample value
1	id (path parameter)	Identifier of the product whose detail is requested.	Positive-integer string, pattern $^[1-9]\d{*}$$	235

8. Output Data

Table B.1 – General output data of product (ProductDetailDto)

No	Attribute	Meaning	Format / Type	Sample value
1	id	Product identifier (numeric primary key returned as string).	String (numeric)	235
2	barcode	Unique code identifying the product.	String	BK10293847
3	title	Product name shown as the main title.	Text	Doraemon Volume 1
4	type	Product type, normalized to upper case.	Enum: BOOK / NEWSPAPER / CD / DVD	BOOK
5	currentPrice	Current selling price.	Number (VND)	85000

6	originalValue	Original / import value (management data). Returned by the API; shown on the manager view.	Number (VND)	70000
7	description	Short product description.	Text	A popular children's comic book
8	weight	Product weight in grams (the UI appends “g”).	Number	350
9	dimensions	Size object with height, width and length.	Object { height, width, length }	{ height: 20, width: 14, length: 2 }
10	imageUrl	Product image URL (optional; may be undefined).	String (optional)	https://.../doraemon_vol1.png
11	stockQuantity	Units currently available in inventory.	Non-negative integer	18
12	status	Product status, normalized. ACTIVE with stockQuantity = 0 is mapped to OUT_OF_STOCK.	Enum: ACTIVE / OUT_OF_STOCK / DEACTIVATED / UNAVAILABLE	ACTIVE
13	specificInfo	Type-specific information object (see Tables B.2–B.5 by type).	Object	see B.2–B.5

Table B.2 – Specific information for Book (BookSpecificInfoDto)

No	Field	Meaning	Example
1	authors	Author(s) of the book.	[“Fujiko F. Fujio”]
2	coverType	Physical cover format of the book.	Paperback
3	publisher	Organization that publishes the book.	Kim Dong Publishing House
4	publicationDate	Publish date (ISO yyyy-mm-dd).	2024-01-15
5	numberOfPages	Total number of pages.	192
6	language	Language of the content.	Vietnamese
7	genre	Literary type / subject category.	Comic

Table B.3 – Specific information for Newspaper (NewspaperSpecificInfoDto)

No	Field	Meaning	Example
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1	editorInChief	Person responsible for editorial management.	Tran Trung Nghia
2	publisher	Organization issuing the newspaper.	Tuoi Tre News
3	publicationDate	Date of issue (ISO yyyy-mm-dd).	2026-03-23
4	issueNumber	Edition number of the newspaper.	No. 369
5	releaseFrequency	How often the newspaper is published.	Daily
6	issn	International standard serial number.	2734-9123
7	language	Main language of the newspaper.	Vietnamese
8	sections	Key content sections of the issue.	["Education", "Sports"]

Table B.4 – Specific information for CD (CDSpecificInfoDto)

No	Field	Meaning	Example
1	artists	Performer(s) or band featured on the CD.	["Westlife"]
2	recordLabel	Company releasing the CD.	Sony Music
3	trackList	Songs included in the album.	["1. My Love – 3:52"]
4	genre	General style of music.	Pop
5	releaseDate	Official release date (ISO yyyy-mm-dd).	2025-09-09

Table B.5 – Specific information for DVD (DVDSpecificInfoDto)

No	Field	Meaning	Example
1	discType	Technical format of the disc.	DVD-9
2	director	Director of the movie / video content.	Hayao Miyazaki
3	runtime	Total running time, in minutes.	169
4	studio	Production / distribution studio.	Studio Ghibli
5	language	Main audio language on the disc.	Japanese
6	subtitles	Available subtitle languages.	["English", "Vietnamese"]
7	releaseDate	Official release date (ISO yyyy-mm-dd).	2023-07-14

8	genre	Category of the movie content.	Animation/Fantasy
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9. Postconditions

- If the request is valid, the system returns and the client displays the detailed information (general + type-specific) of the selected product.
- The use case is read-only: no business data is created, updated, or deleted.
- On failure, the system returns the corresponding HTTP status (400 for an invalid id, 404 when the product is not found, 500 for a database/service error) and the client shows an appropriate message; the actor can retry or go back. This use case does not write a dedicated error-log entry.